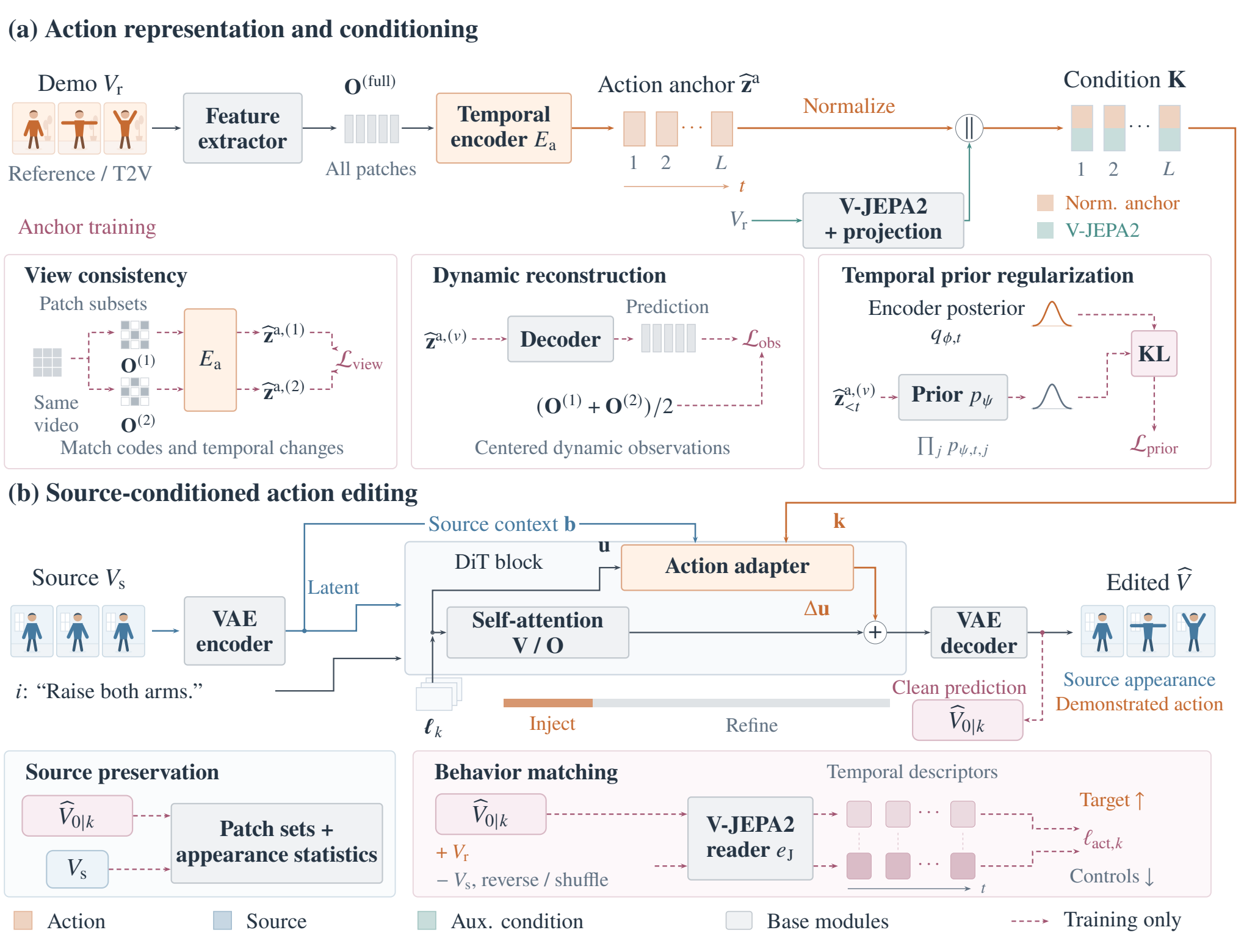

(a) Action representation and conditioning
Demo V_r
Reference / T2V
Feature extractor
$\mathbf{O}^{(full)}$
All patches
Temporal encoder E_a
Action anchor $\hat{\mathbf{z}}^a$
1 2 L
t
Normalize
Condition K
1 2 L
Norm. anchor
V-JEPA2
V_r
V-JEPA2 + projection
Anchor training
View consistency
Patch subsets
Same video
$\mathbf{O}^{(1)}$
$\mathbf{O}^{(2)}$
E_a
$\hat{\mathbf{z}}^{a,(1)}$
$\hat{\mathbf{z}}^{a,(2)}$
$\mathcal{L}_{view}$
Match codes and temporal changes
Dynamic reconstruction
Prediction
$\hat{\mathbf{z}}^{a,(v)}$
Decoder
$\mathcal{L}_{obs}$
$(\mathbf{O}^{(1)} + \mathbf{O}^{(2)})/2$
Centered dynamic observations
Temporal prior regularization
Encoder posterior
$q_{\phi,t}$
KL
$\hat{\mathbf{z}}^{a,(v)}_{<t}$
Prior p_ψ
$\prod_j p_{\psi,t,j}$
$\mathcal{L}_{prior}$
(b) Source-conditioned action editing
Source context b
k
DiT block
u
Action adapter
Δu
Source V_s
VAE encoder
Latent
Self-attention V / O
VAE decoder
Edited $\hat{V}$
i: "Raise both arms."
ℓ_k
Inject
Refine
Clean prediction
$\hat{V}_{0|k}$
Source appearance
Demonstrated action
Source preservation
$\hat{V}_{0|k}$
V_s
Patch sets + appearance statistics
Behavior matching
Temporal descriptors
$\hat{V}_{0|k}$
+ V_r
− V_s, reverse / shuffle
V-JEPA2 reader e_J
Target ↑
$\ell_{act,k}$
Controls ↓
t
Action
Source
Aux. condition
Base modules
Training only